

# Modelling of Stiffness

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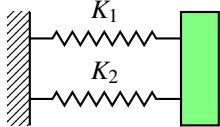
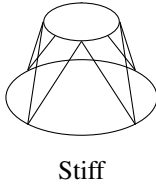
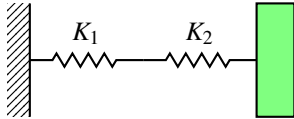
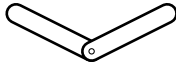
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## 1 Intro

We're going to talk about how to model the flexibility of a joint. Real mechanisms are never infinitely stiff, although with many robotic mechanisms the stiffness is high enough that we can mostly ignore it, or at least design suitable control laws such that we don't excite flex modes. In some cases, however, that isn't possible. Very long robot arms, such as the SRMS and SSRMS, have nontrivial flex dynamics that need to be modeled and managed, either via appropriate control laws or via operational procedures. Also, there is emerging research in soft robotics where the flexibility of a manipulator is in fact one of its defining features.

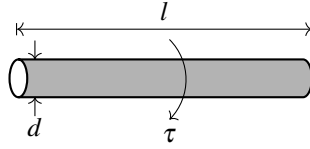
That being said, this isn't going to be a lecture so much as it is a summary of the types of flexible structures and mechanisms you might encounter in the wild and accepted equations for modeling them. We aren't going to derive or even really motivate these models; the modeling of deformable structures is at least an entire graduate-level class.

One introductory note: if you have two or more structural elements in combination, you can find the flexibility of the assembly as follows.

| Type     | Diagram                                                                             | Example                                                                                           | Equation                                                   |
|----------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Parallel |  | <br>Stiff     | $K_{\text{par}} = K_1 + K_2$                               |
| Series   |  | <br>Not stiff | $\frac{1}{K_{\text{ser}}} = \frac{1}{K_1} + \frac{1}{K_2}$ |

## 2 Flexible Elements

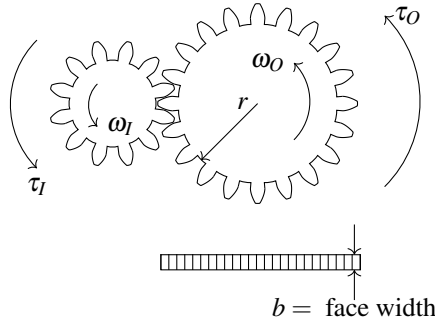
Shaft



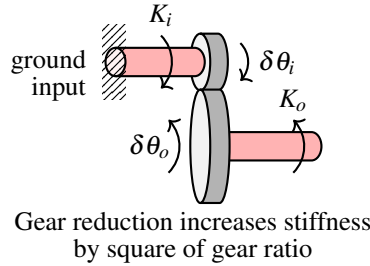
$$K = \frac{G\pi d^4}{32l}$$

where  $G$  is the “shear modulus”  
(steel =  $7.5 \times 10^{10}$  N/m<sup>2</sup>)

Gear

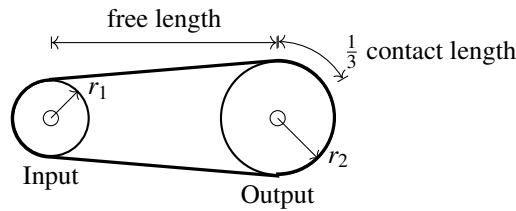


$$\begin{aligned} r &= \text{radius of output gear} \\ \eta &= \text{gear ratio} \\ \omega_o &= \frac{\omega_i}{\eta} \\ \tau_o &= \tau_i \eta \\ K &= C_g b r^2 \text{ (locked input gear)} \\ C_g &= 1.34 \times 10^{10} \text{ N/m}^2 \text{ (steel)} \end{aligned}$$



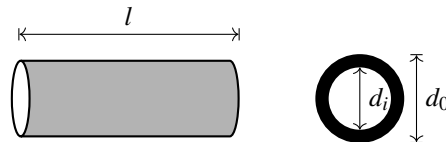
$$\begin{aligned} \tau_i &= K_i \delta \theta_i \\ \tau_o &= K_o \delta \theta_o \\ K_o &= \frac{\tau_o}{\delta \theta_o} = \frac{\eta \tau_i}{\delta \theta_i / \eta} \\ &= \frac{\eta K_i \delta \theta_i}{\delta \theta_i / \eta} = \eta^2 K_i \end{aligned}$$

Belt

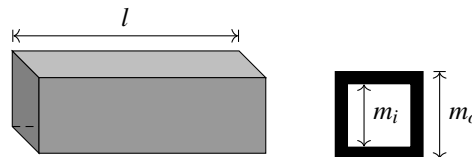


$$\begin{aligned} K &= \frac{AE}{l} \\ A &= \text{cross sectional area of belt} \\ E &= \text{modulus of elasticity} \\ l &= \text{free length} + \frac{1}{3} \text{ contact length} \end{aligned}$$

Link



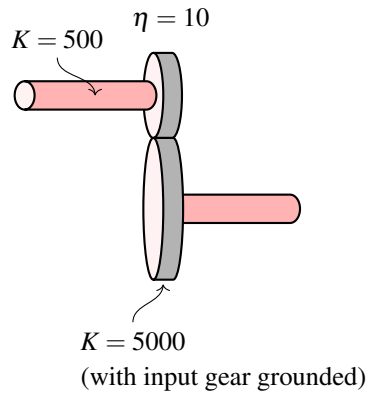
$$\begin{aligned} K &= \frac{3\pi E(d_o^4 - d_i^4)}{64l^3} \\ E &= 2 \times 10^{11} \text{ N/m}^2 \text{ (steel)} \\ &= \frac{2}{3} \times 10^{11} \text{ N/m}^2 \text{ (aluminum)} \end{aligned}$$



$$K = \frac{E(w_o^4 - w_i^4)}{4l^3}$$

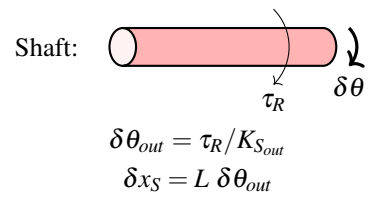
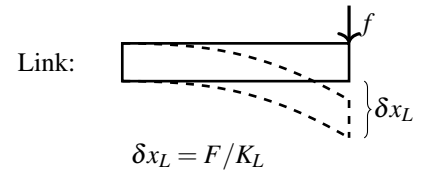
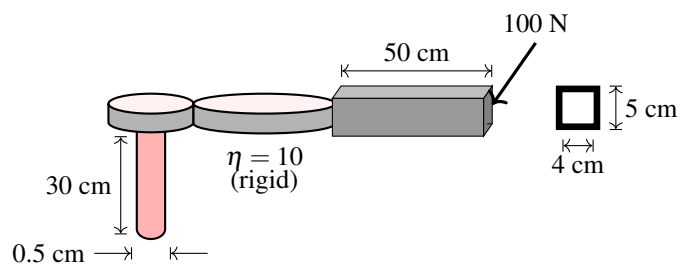
### 3 Example: Gear Stiffness

What is the output stiffness of the drive system below?



$$\begin{aligned}\frac{1}{K_{sys}} &= \frac{1}{K_{gear}} + \frac{1}{K_{shaft}} \\ &= \underbrace{\frac{1}{500}}_{\text{at output}} + \underbrace{\frac{1}{(10)^2(500)}}_{50000} \\ K_{sys} &= 4545 \text{ N-m/rad}\end{aligned}$$

### 4 Example



$$K_{shaft} = \frac{G\pi d^4}{32l} = \frac{(7.5 \times 10^{10})(\pi)(5 \times 10^{-3})^4}{32(0.3)} = 15.3 \text{ N-m/rad} \rightarrow K_{S_{out}} = 15.3(10)^2 = 1530 \text{ N-m/rad}$$

$$K_{link} = \frac{E(w_o^4 - w_i^4)}{4l^3} = \frac{(2 \times 10^{11})(0.05^4 - 0.04^4)}{4(0.5)^3} = 1.476 \times 10^{-5} \text{ N/m}$$

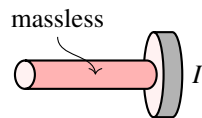
Link:  $\delta x_L = 100 \text{ N} / 1.476 \times 10^6 \text{ N-m} = 6.78 \times 10^{-5} \text{ N/m}$

Shaft:  $\delta x_S = (0.50 \text{ m})(0.5 \text{ m} \times 100 \text{ N}) / 1530 \text{ N-m/rad} = 0.0163 \text{ N-m/rad}$

## 5 Estimating $\omega_r$

$$\omega_n = \sqrt{\frac{k}{m}} \begin{cases} \rightarrow \text{stiffness} \\ \rightarrow \text{effective mass or inertia} \end{cases}$$

### 5.1 Example: Shaft

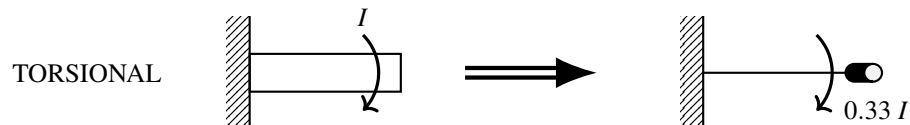
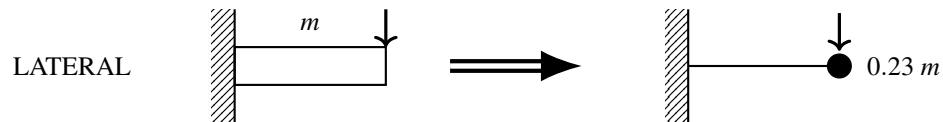


$$k = 400 \text{ N-m/rad}$$

$$I = 1 \text{ kg-m}^2$$

$$\omega_r = \sqrt{\frac{400}{m}} = 20 \text{ rad/sec} = 3.2 \text{ Hz}$$

### 5.2 Lumped mass model



### 5.3 Example: 9.9: Link

$$\begin{aligned}m &= 4.347 \text{ kg} \\k &= 3600 \text{ N/m}\end{aligned}$$

What is  $\omega_r$ ?

Assume mass is distributed. Then,

$$\begin{aligned}m_{\text{eff}} &= 0.23(4.345 \text{ kg}) = 1 \text{ kg} \\ \omega_r &= \sqrt{\frac{k}{m_{\text{eff}}}} = \sqrt{\frac{3600}{1}} = 60 \text{ rad/sec} = 9.6 \text{ Hz}\end{aligned}$$